Supplementary materials

An OpenFOAM-Based Solver for Three-Phase Flow in Porous Media Incorporating Foam Effects

Section 1 presents validation studies for two and three-phase flow under gravitational effects, based on the works of Durlofsky [9], Abreu [1], and Abreu et al. [2], considering different ratios of gravitational to convective forces. Section 2 provides validation cases addressing capillary pressure effects. It first reproduces the two-phase capillary–gravity equilibrium problem of Horgue et al. [11] and Carrillo et al. [5], solving for the steady-state saturation profile in a one-dimensional water–air column, and subsequently it presents a analysis of a three-phase flow case using the Leverett model for capillary pressure [12] with varying capillary forces, based on the work of Abreu et al. [2]. Section 3 presents a study case based on the laboratory five-spot waterflood experiment of Gaucher and Lindley [10], validating the implementation with respect to source and sink terms under scaled laboratory conditions. The present section focuses on the validation of multiphase foam flow. Together, all the selected validation cases encompass classical reference benchmarks and experimental observations, thereby providing a comprehensive assessment of ImpesFOAM performance under different flow conditions.

1. Benchmark case – Gravitational effects

1.1. Two-phase flow

The validation study in this section follows the work of Durlofsky [9] and de Abreu [8], presenting results for simple one-dimensional water–oil displacement cases ($S_w + S_o = 1$). Firstly, a dimensionless number, G_d, is introduced to quantify the ratio of gravitational to convective effects, defined as:

$$G_d = \frac{k_c \Delta \rho g}{\mu_o \nu_c}, \tag{1}$$

where k_c is the characteristic permeability (e.g., the geometric average of the element permeabilities), ν_c is the characteristic total velocity, $\Delta\rho = \rho_w - \rho_o$, g the gravitational acceleration, and μ_o the dynamic viscosity of the oil.

For the numerical simulations of one-dimensional two-phase flow presented in this section, the following Riemann problem (RP) is considered, where the left and right states are given by:

$$RP = \begin{cases} S_w^L = 1.0 \quad \text{and} \quad S_w^R = 0, \\ S_o^L = 0 \quad\;\;\, \text{and} \quad S_o^R = 1.0. \end{cases} \tag{2}$$

The RP defined in Eq. (2) represents a domain initially saturated with oil (right state, R) into which only water is injected (left state, L).

The classical quadratic Corey-type model [7] was used as the relative permeability functions, as follows:

$$k_{rw} = S_w^2 \quad \text{and} \quad k_{ro} = S_o^2 = (1 - S_w)^2. \tag{3}$$

For defining the dimensionless number (G_d) in the two tested scenarios, the parameters listed in Table 1 were used. Additionally, the ratio of the dynamic viscosities of the fluids was assumed to be $\mu_o/\mu_w = 5$. The first simulation corresponds to $G_d = 0$ (i.e., no gravity effects, $g = 0$), and the second to $G_d = 2$. The value of μ_o is determined using Eq. (1) by isolating μ_o, and μ_w is then calculated based on the assumption $\mu_o/\mu_w = 5$.

The dimensionless time t_D is evaluated as:

$$t_D = \frac{v_c \cdot t}{(1 - S_{wc} - S_{or})\, L\phi}, \tag{4}$$

Parameter	Value	Unit	Parameter	Value	Unit
ρ_o	0.7	kg/m^3	ρ_w	1.0	kg/m^3
g	9.81	m/s^2	v_c	1.0	m/s
k_c	1.0×10^{-13}	m^2	ϕ	0.25	–
S_{wc}	0	–	S_{ro}	0	–
L	1.0	m	N_x	100	–
Δx	0.01	m	t_{final}	0.125	s
Δt	1×10^{-5}	s			

Table 1: Simulation parameters for the two-phase gravity effects benchmark.

where L is the domain length and ϕ the porosity. Figure 1 shows the our numerical results (ImpesFOAM), which present a good agreement with the literature [9] and with the analytical solution using the fractional flow theory, considering $f_w = \frac{\lambda_w}{\lambda_t} + \frac{K}{q_t}\frac{\lambda_w \lambda_o}{\lambda_t}(\rho_w - \rho_o)g$ [3].

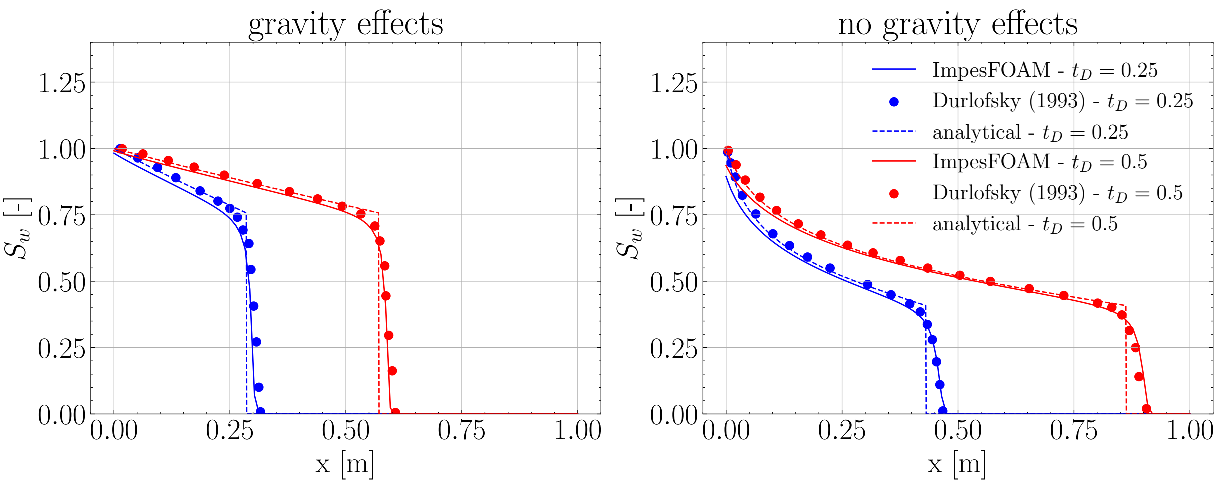


Figure 1: Solution profiles for one-dimensional displacement with $G_d = 2$ (left) and $G_d = 0$ (right) comparing the results from analytical solution [3], numerical results: ImpesFoam (proposed here) and by Durlofsky [9].

Additionally, we compared in Fig. 2 the accuracy of the results when using different interpolation schemes available in OpenFOAM framework. One can observe that the van-Leer [16] method achieved the best results, so it will be considered in subsequent simulations.

1.2. Three-phase flow

For the three-phase flow, we first present numerical simulations of one-dimensional flow using two Riemann problems proposed by Abreu et al. [2] for a three-phase system neglecting gravity and capillary pressure effects. Section 2.2 examine capillary pressure effects on both Riemann problems [2], where the Leverett model [12] is employed for capillary pressure. Conversely, this section examine gravity effects on RP_2 [1]. The Riemam Problems are given by:

$$RP_1 = \begin{cases} S_w^L = 0.613 \quad \text{and} \quad S_w^R = 0.05, \\ S_g^L = 0.387 \quad \text{and} \quad S_g^R = 0.15. \end{cases} \tag{5}$$

$$RP_2 = \begin{cases} S_w^L = 0.721 \quad \text{and} \quad S_w^R = 0.05, \\ S_g^L = 0.279 \quad \text{and} \quad S_g^R = 0.15. \end{cases} \tag{6}$$

The classical quadratic Corey-type model [2] was used as the relative permeability functions, as follows:

$$k_{rw} = S_w^2, \quad k_{rg} = S_g^2 \quad \text{and} \quad k_{ro} = S_o^2 = (1 - S_w - Sg)^2. \tag{7}$$

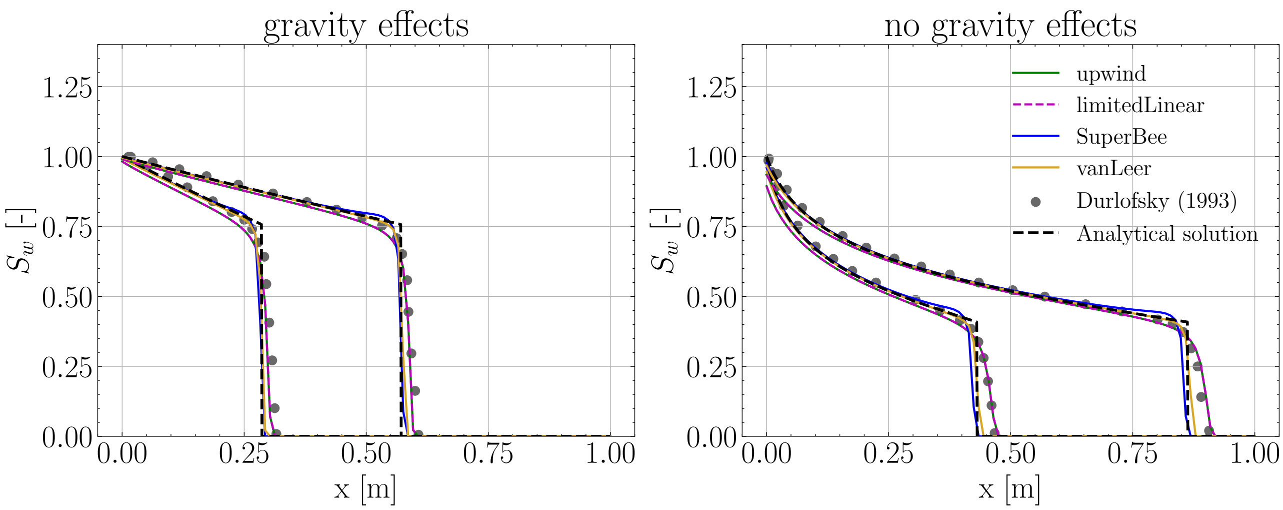


Figure 2: Comparison of the results using different interpolation schemes available in OpenFOAM with those from literature [9] and analytical solution following the methodology outlined by Blunt [3].

The parameters employed in the simulations are listed in Table 2. Figure 3 presents the phase saturation profiles at two different time instants, compared with the reference results of Abreu et al. [2]. The results show good agreement with almost indistinguishable profiles for phases saturation.

Parameter	**Value**	**Unit**	**Parameter**	**Value**	**Unit**
ρ_g	5.76×10^{-2}	kg/m^3	ρ_o	0.7	kg/m^3
ρ_w	1.0	kg/m^3	μ_g	0.3	cP
μ_o	1.0	cP	μ_w	0.5	cP
v_c	1.0	m/s	ϕ	1.0	–
$S_{wc} = S_{ro} = S_{gr}$	0	–	$n_w = n_o = n_g$	2.0	–
L_x	0.051	m	N_x	1000	–
Δx	5.1×10^{-5}	m	t_{final}	150	s
Δt	1.0×10^{-4}	s			

Table 2: Simulation parameters for three-phase flow benchmark.

Next we conduct a case study based on the RP_2 defined in Eq. (6), now accounting for gravitational effects [1]. The dimensionless group G_d was evaluated using the parameters listed in Table 2. Two scenarios are considered: the first corresponds to $G_d = 0$ (i.e., no gravity effects, $g = 0$), which was already presented in Fig. 3; and the second corresponds to $G_d = 0.0475$, representing a minor influence of gravity. The characteristic permeability k_c is computed from Eq. (1) as a function of the prescribed G_d value.

Figure 4 presents the saturation profiles for the three phases at a specific time instant, without diffusion, i.e., no capillary pressure, illustrating the agreement between the results obtained from ImpesFOAM solver and those reported by Abreu [1]. It can be seem as a slightly perturbation induced by the dimensionless group $G_d = 0.0475$ (ratio of gravity to convection). The results illustrate the influence of the gravity, which induces a slight perturbation. Both simulations show that the gas saturation front under gravity (blue line) advances slightly ahead compared to the case without gravity (black dashed line). Also, it is observed that the plateau achieved in the gas saturation around $S_g \approx 0.215$ is slightly lower than the one obtained without gravity effects, $S_g \approx 0.219$. The reference case without gravity, $G_d = 0$, is the same profile previously shown in Fig. 3 and is replotted here for comparison. These results are consistent with the expected effect of gravity acting in the opposite direction of the injection direction [1].

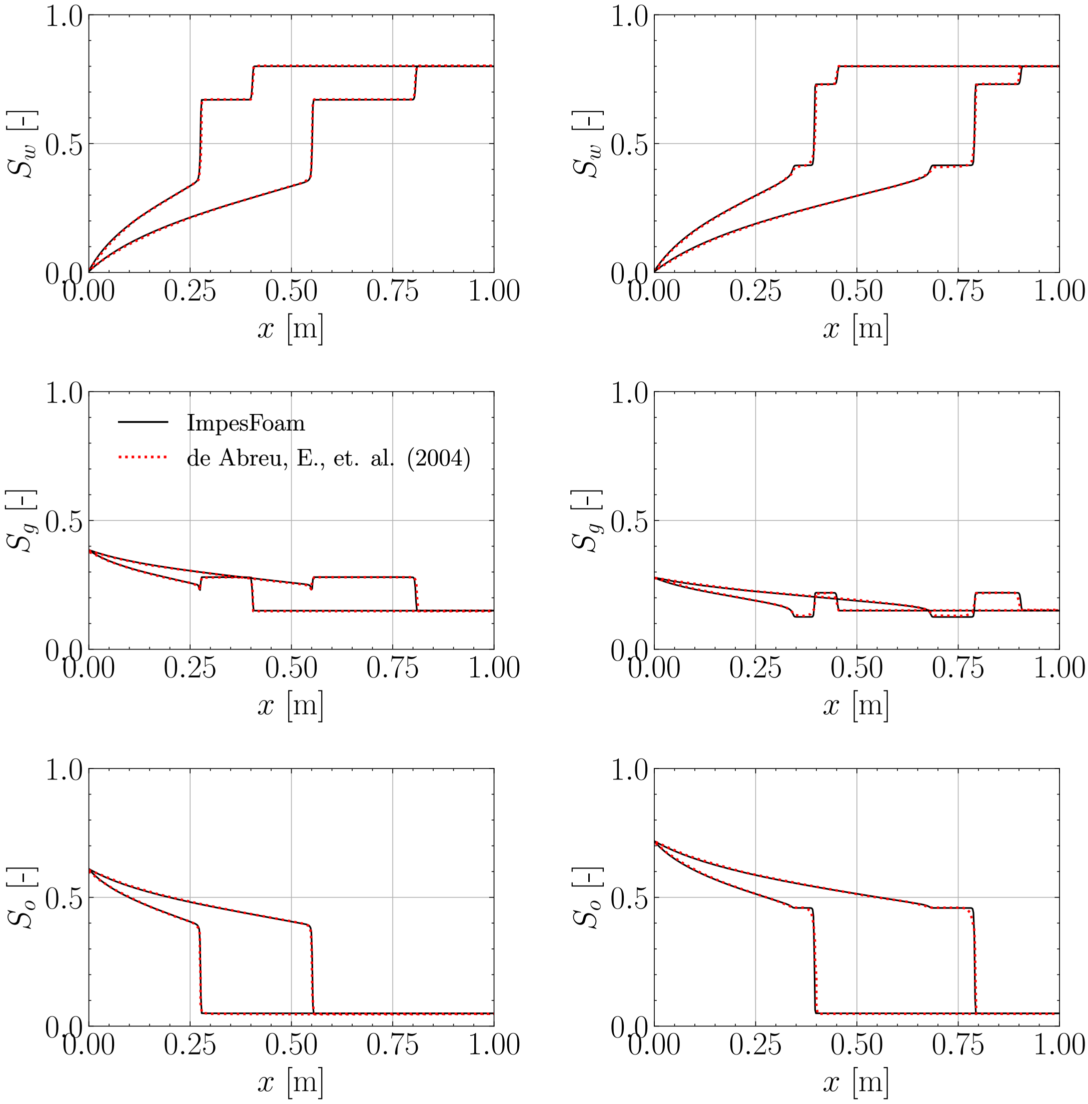


Figure 3: Comparison of one-dimensional displacement profiles obtained with ImpesFOAM against the reference solutions of Abreu et al. [2], for the case without gravity and capillary-pressure effects.

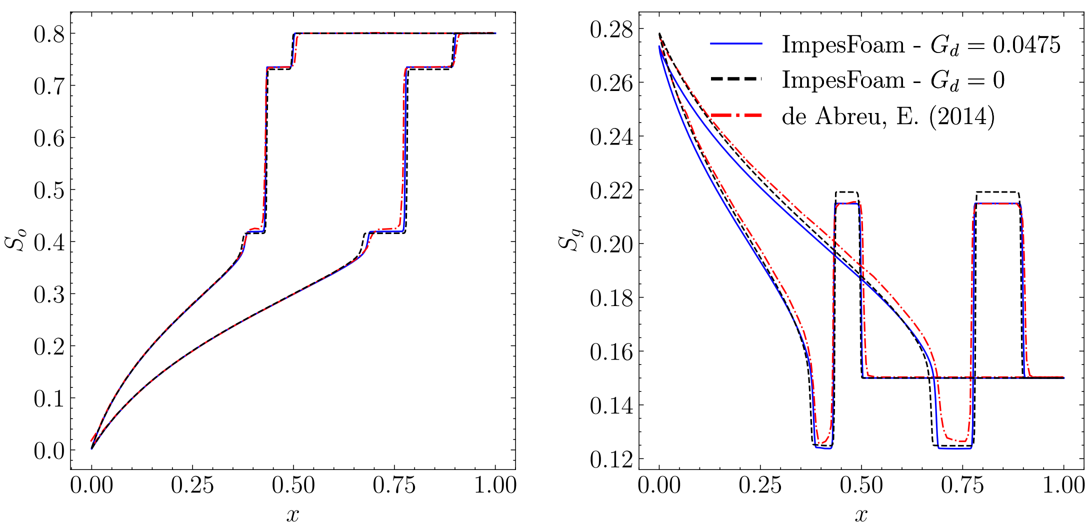


Figure 4: Comparison of solution profiles for one-dimensional displacement with $G_d = 0.0475$ between ImpesFOAM and results from Abreu [1].

2. Benchmark case – Capillary pressure effects

2.1. Two-phase capillary pressure and gravity equilibrium

Now we present a validation case based on the work of Horgue et al. [11] and Carrillo et al. [5] modeling the capillary-gravity equilibrium by solving for the steady state saturation profile of a one-dimensional porous column filled with water and air, as depicted in Fig. 5. Here, the initial water saturation of the column is set far from its thermodynamic equilibrium in a step-wise fashion: the lower half is partially saturated with water ($S_w = 0.5$) while the upper half is initially dry as shown in Fig. 5. To ensure proper equilibriation, both fluids are allowed to flow freely through the column's top boundary, but not through its lower one, as depicted by the boundary conditions shown in Fig. 5. For this simplified case, the theoretical steady-state can be described as the balance between capillary and gravitational forces, where gravity pulls the heavier fluid (water) downwards while capillarity pulls it upwards, which is described by [11, 5]:

$$\frac{\partial p_c}{\partial x} = (\rho_g - \rho_w)\, g, \tag{8}$$

which can be rearranged using the chain rule to yield:

$$\frac{\partial S_{w,pc}}{\partial x} = \frac{(\rho_g - \rho_w)\, g}{\frac{\partial p_c(S_{w,pc})}{\partial S_{w,pc}}}. \tag{9}$$

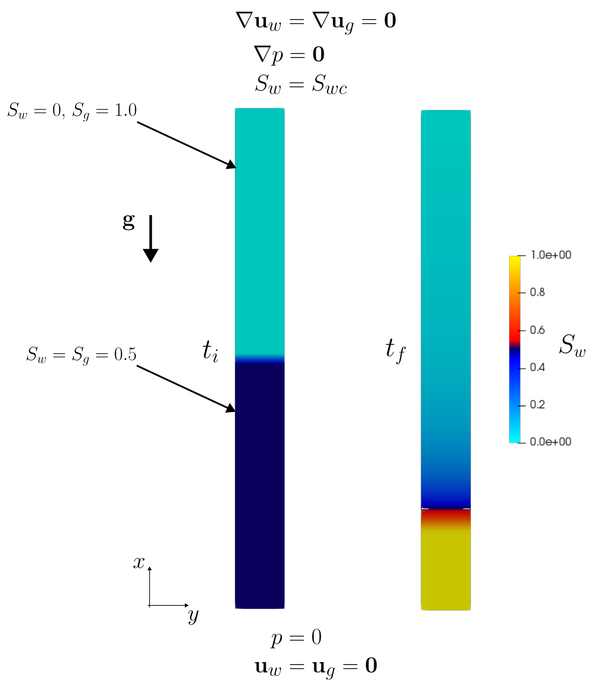


Figure 5: Schematic of the initial and boundary conditions used in the simulations. The figure shows the initial water saturation, S_w, distribution with which the porous column is saturated, t_i (left). The steady-state configuration of the domain is also illustrated at the final simulation time, t_f (left). The imposed lower and upper boundary conditions are indicated, as well as the direction of the gravity vector $\mathbf{g}$. Adapted from Carrillo et al. [5].

This steady-state saturation profile and $\frac{\partial S_{w,pc}}{\partial x}$ at steady-state are used as reference results. The capillary pressure correlations are based on the notion of effective saturation. However, the macro-scale capillary pressure tends to infinity when saturation S_w tends to S_{wc}. To accept irreducible and maximal saturations in numerical simulations, we define the effective saturation for capillary pressure $S_{w,pc}$ as follows [11]:

$$S_{w,pc} = \frac{S_w - S_{pc,irr}}{S_{pc,max} - S_{pc,irr}} \tag{10}$$

where the minimal $S_{pc,irr}$ is a user-defined parameter that should satisfy $S_{pc,irr} < S_{wc}$. The correlation for capillary pressure proposed by Brooks and Corey [4] is given by:

$$p_c\left(S_{w,pc}\right) = p_{c,0} S_{w,pc}^{-\alpha} \tag{11}$$

where $p_{c,0}$ is the entry capillary pressure and $1/\alpha$ the pore-size distribution index. Deriving Eq. (11), $\frac{\partial p_c}{\partial S_b}$ can be expressed as:

$$\frac{\partial p_c}{\partial S_{w,pc}}\left(S_{w,pc}\right) = -\frac{\alpha p_{c,0}}{S_{pc,max} - S_{pc,irr}} S_{w,pc}^{-\alpha-1} \tag{12}$$

The $p_c\left(S_{w,pc}\right)$ and $\frac{\partial p_c}{\partial S_{w,pc}}\left(S_{w,pc}\right)$ curves are show in Fig. 6 using parameters from Table 3, which were used in the numerical simulations.

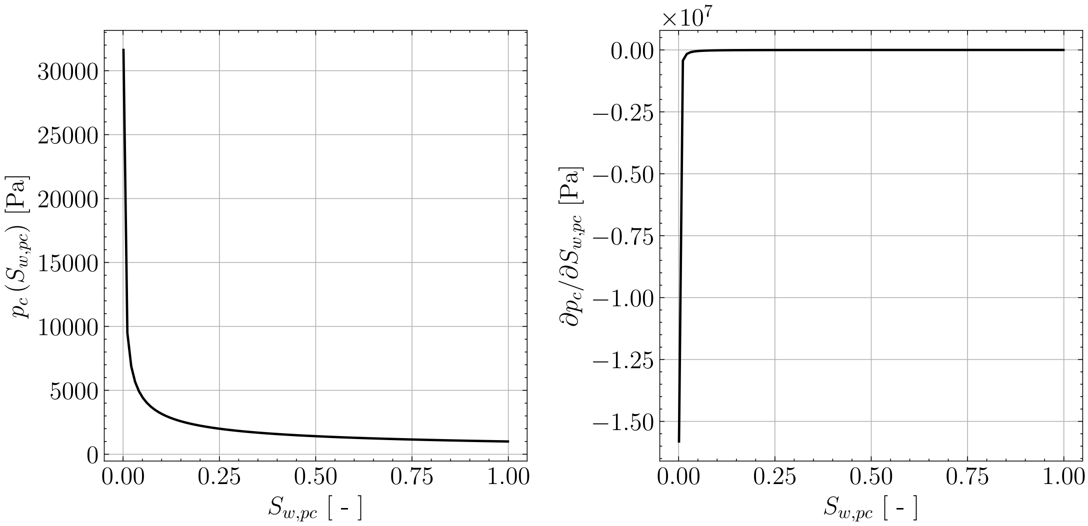


Figure 6: Capillary pressure $p_c\left(S_{w,pc}\right)$ curve (left) and its derivative $\frac{\partial p_c}{\partial S_{w,pc}}\left(S_{w,pc}\right)$ (right) using parameters from Table 3.

Parameter	**Value**	**Unit**	**Parameter**	**Value**	**Unit**
ρ_g	1.0	kg/m^3	ρ_w	1000	kg/m^3
μ_g	1.76×10^{-5}	Pa·s	μ_w	1.0×10^{-3}	Pa·s
g	9.81	m/s^2	ϕ	0.5	–
S_{wc}	1.0×10^{-3}	–	S_{gr}	1.0×10^{-3}	–
n_w	3.0	–	n_g	3.0	–
k_{rw}^0	1.0	–	k_{rg}^0	1.0	–
L	1.0	m	N_x	100	–
Δx	0.01	m	t_{final}	2.0×10^6	s
Δt	10.0	s			

Table 3: Simulation parameters for capillary pressure and gravity equilibrium benchmark.

Figure 7 compares the numerical results obtained with the Corey–Brooks relative permeability model against the analytical solution and other numerical results reported in the literature [5].

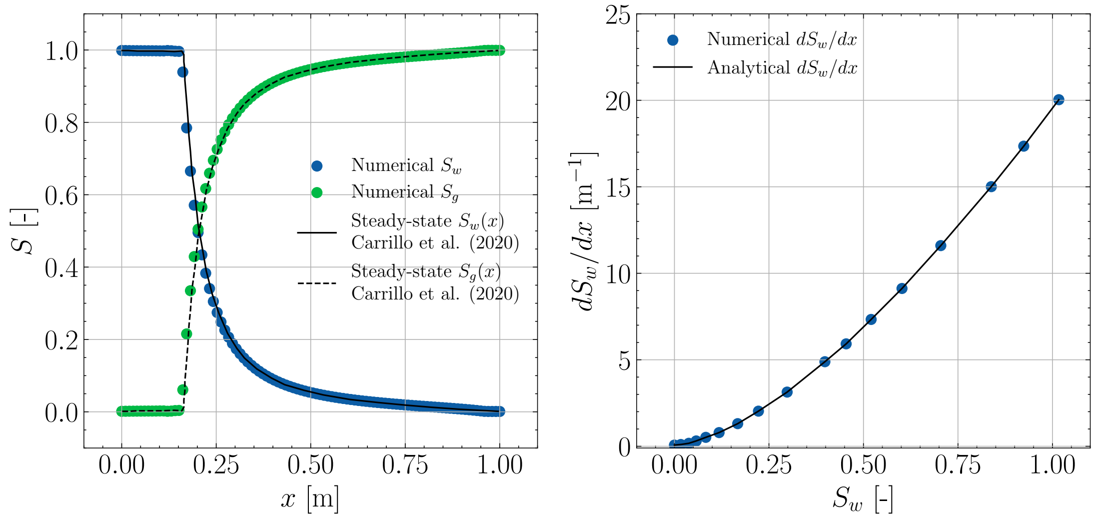


Figure 7: Comparison of steady-state water saturation profiles from our numerical (dots) framework with the analytical solution in Eq. (9) and numerical results from literature [5]. The left panel shows the steady-state saturation profiles, while the right panel presents the corresponding equilibrium water saturation gradient for the Corey–Brooks capillary pressure model [4].

2.2. Three-phase flow with capillary pressure effects

Finally, we analyze RP_1 and RP_2 cases using the Leverett model for capillary pressure [12], while neglecting gravity effects. The Leverett curves are given by:

$$p_{cwo} = 5\epsilon(2 - S_w)(1 - S_w) \qquad \text{and} \qquad p_{cog} = 5\epsilon(2 - S_g)(1 - S_g), \tag{13}$$

where ϵ represents the relative importance of advective and capillary/dispersive forces. It is defined as the inverse of Péclet number (Pe):

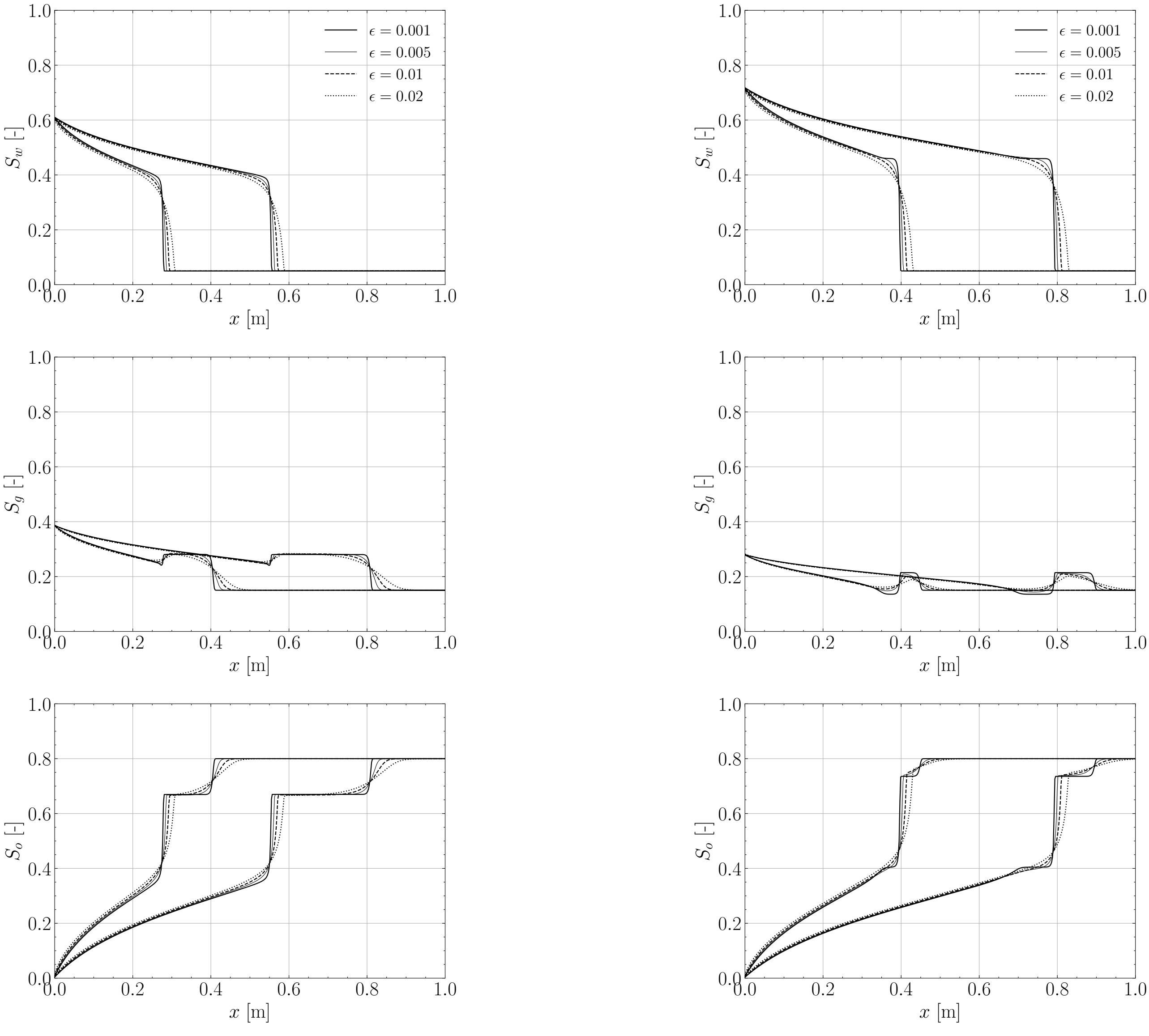
$$\epsilon = \frac{1}{Pe}, \qquad Pe = \frac{\nu_c \mu_o L}{p_c k_c}, \tag{14}$$

where k_c is the characteristic permeability, ν_c is the characteristic total velocity, μ_o the dynamic viscosity of the oil, L is the maximum characteristic length of the system, and p_c is the characteristic value for capillary pressures. Following the procedure adopted by Abreu et al. [2] to assess the influence of diffusive effects induced by capillary pressure, the parameter ϵ is varied over the values 0.001, 0.005, 0.01, and 0.02.

Figure 8 presents the solutions obtained on a grid with 500 cells. A clear qualitative resemblance can be observed between the solutions of RP_1 and RP_2 and those previously shown in Fig. 3. The main difference lies in the diffusive behavior, which becomes more pronounced as ϵ increases. This trend is consistent with the observations reported by Abreu et al. [2].

3. Benchmark case – Five-spot waterflood

Next we present a study case based on the laboratory 5-spot water flood described by Gaucher and Lindley [10], and hence also serves to validate the model with respect to source and sink as well as with scaled laboratory experiments. The 5-spot well pattern is depicted in Fig. 9. Water is injected into the injection wells (blue) to displace the resident oil towards the production wells (red). The parameters used for the simulations are shown in Table 5. Each well node was assigned only one-quarter of the given well fluxes due to symmetric considerations [13]. It is important to emphasize that the work of Panday et al. [13] only provides the values of relative permeabilities (k_{rw} and k_{ro}) data versus S_w data, as well as the residual saturations and endpoints. So it is necessary to fit the Corey-Brooks relative permeability exponents (n_w and



(a) RP_1

(b) RP_2

Figure 8: Solutions of the Riemann problems with capillary pressure effects for different values of ϵ.

n_o) using the reported data. The same holds for capillary pressure p_c versus S_w data. Figure 10 shows the data and the model with fitted parameters for relative permeability and capillary pressure functions.

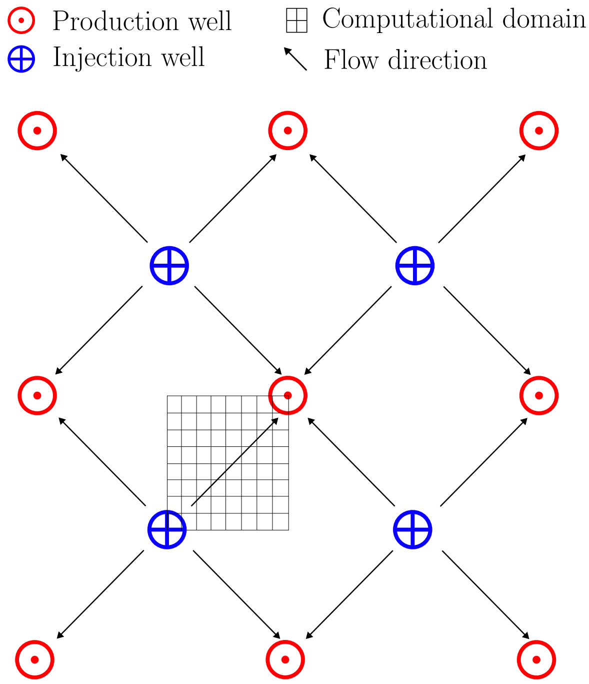


Figure 9: Schematic of the 5-spot water-flood problem. Adapted from [13, 10].

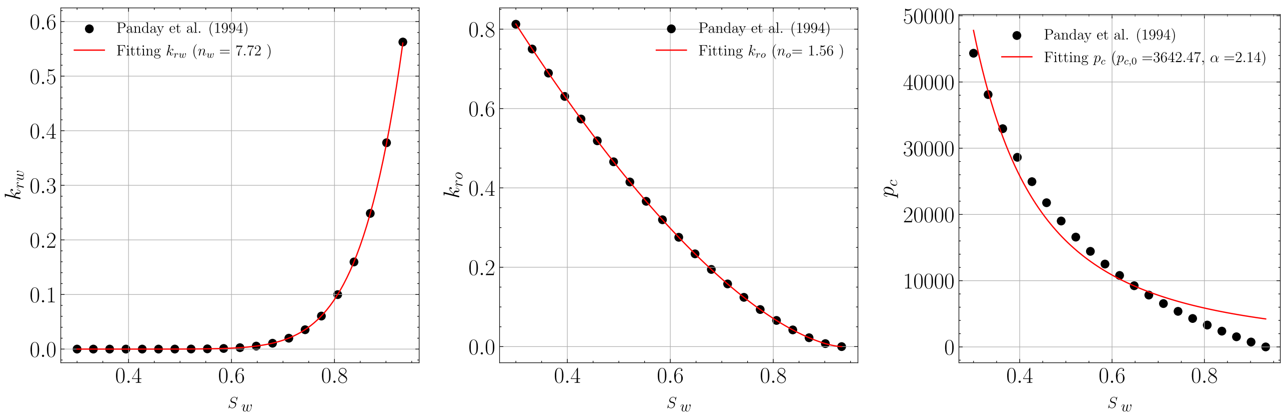


Figure 10: Fitting of n_w, n_o, $p_{c,0}$ and α from data [13].

In the scope of this work we do not focus on the wellbore modeling in porous media, considering only constant injection and extraction flow rates of wellbores. The reader is referred to the work of Peaceman [14], Chen et al. [6] and Quinelato et al. [15] for further details on this topic. It is important to note that the total flow rate q_t of all wells, both producers and injectors, is prescribed at all time steps. For injection wells, we have $q_t = q_w$ and $q_o = 0$, while for production wells, $q_t = q_w + q_o$. Thus, q_w is known at injection wells but not at production wells. In production wells, the water flow rate q_w is related to the total flow rate through a fractional flow function $f(S_w)$, such that: $q_w = f(S_w)q_t$, where $f(S_w)$ denotes the water fractional flow as a function of saturation S_w. In other words, q_w represents a fraction of the total flow rate q_t. Table 4 shows the values for q_w and q_o for both injection and production wells.

The simulation parameters are summarized in Table 5. Figure 11 presents the cumulative oil recovery, comparing experimental data from Gaucher and Lindley [10], numerical results from Panday et al. [13] obtained with finite-difference and finite-element methods, and the results

Wells	Water	Oil	Units
Injection	q_t	0	m^3 / s · $\mathrm{m}^3 \rightarrow 1/\mathrm{s}$
Production	$f_w \cdot qt$	$(1 - f_w) \cdot qt$	m^3 / s · $\mathrm{m}^3 \rightarrow 1/\mathrm{s}$

Table 4: Injection and production wells.

of our simulations. The comparison indicates good agreement with literature. Finally, Fig. 12 illustrates the spatial distribution of water saturation at four time instants, highlighting the displacement dynamics within the domain.

Parameter	**Value**	**Unit**	**Parameter**	**Value**	**Unit**
ρ_o	800	$\mathrm{kg/m^3}$	ρ_w	1000	$\mathrm{kg/m^3}$
μ_o	2.17×10^{-3}	Pa·s	μ_w	0.5×10^{-3}	Pa·s
ϕ	0.2	–	S_{wc}	0.3	–
S_{or}	0.067	–	n_w	7.72 (fitted)	–
n_o	1.56 (fitted)	–	k^0_{rw}	0.562	–
k^0_{ro}	0.813	–	$p_{c,0}$	3642.47 (fitted)	Pa
α	2.14 (fitted)	–	S_{o_i}	0.7	–
q_t	3.1868×10^{-6}	1/s	t_{final}	2.0×10^6	s
$L_x = L_y$	142.245	m	$N_x = N_y$	50	–
$\Delta x = \Delta y$	≈ 2.85	m	t_{final}	5.0×10^8	s
Δt	1.0×10^4	s			

Table 5: Simulation parameters for the 5-spots benchmark.

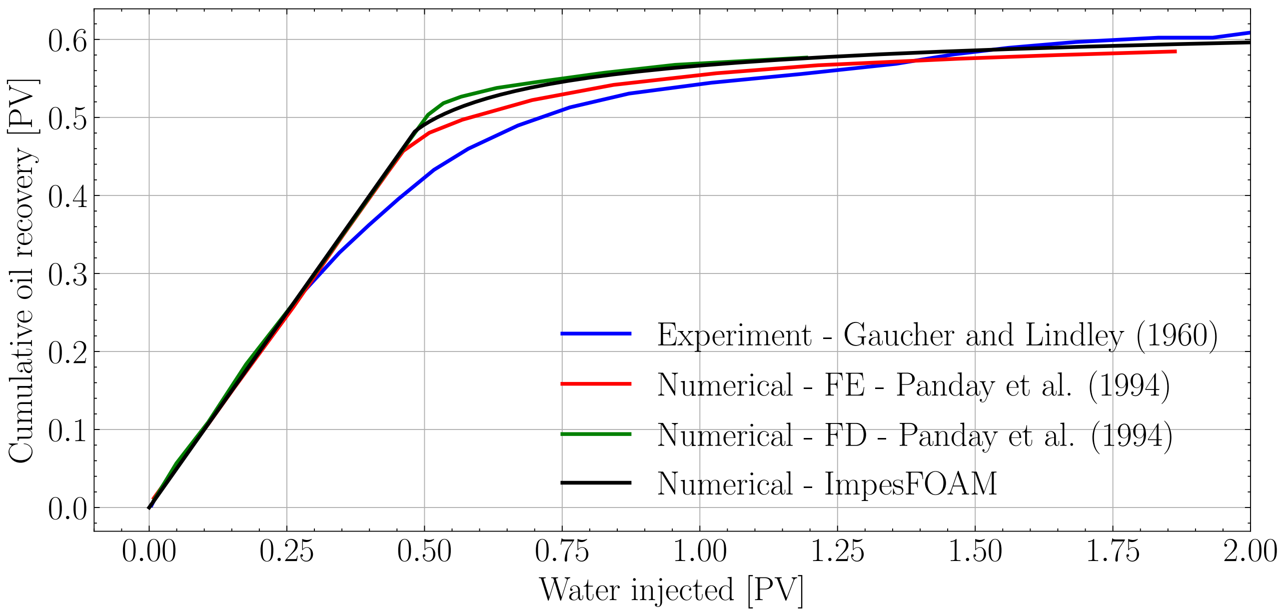


Figure 11: Comparison of cumulative oil recovery between the numerical results obtained in this work, those reported by Panday et al. [13], and the experimental data from Gaucher and Lindley [10].

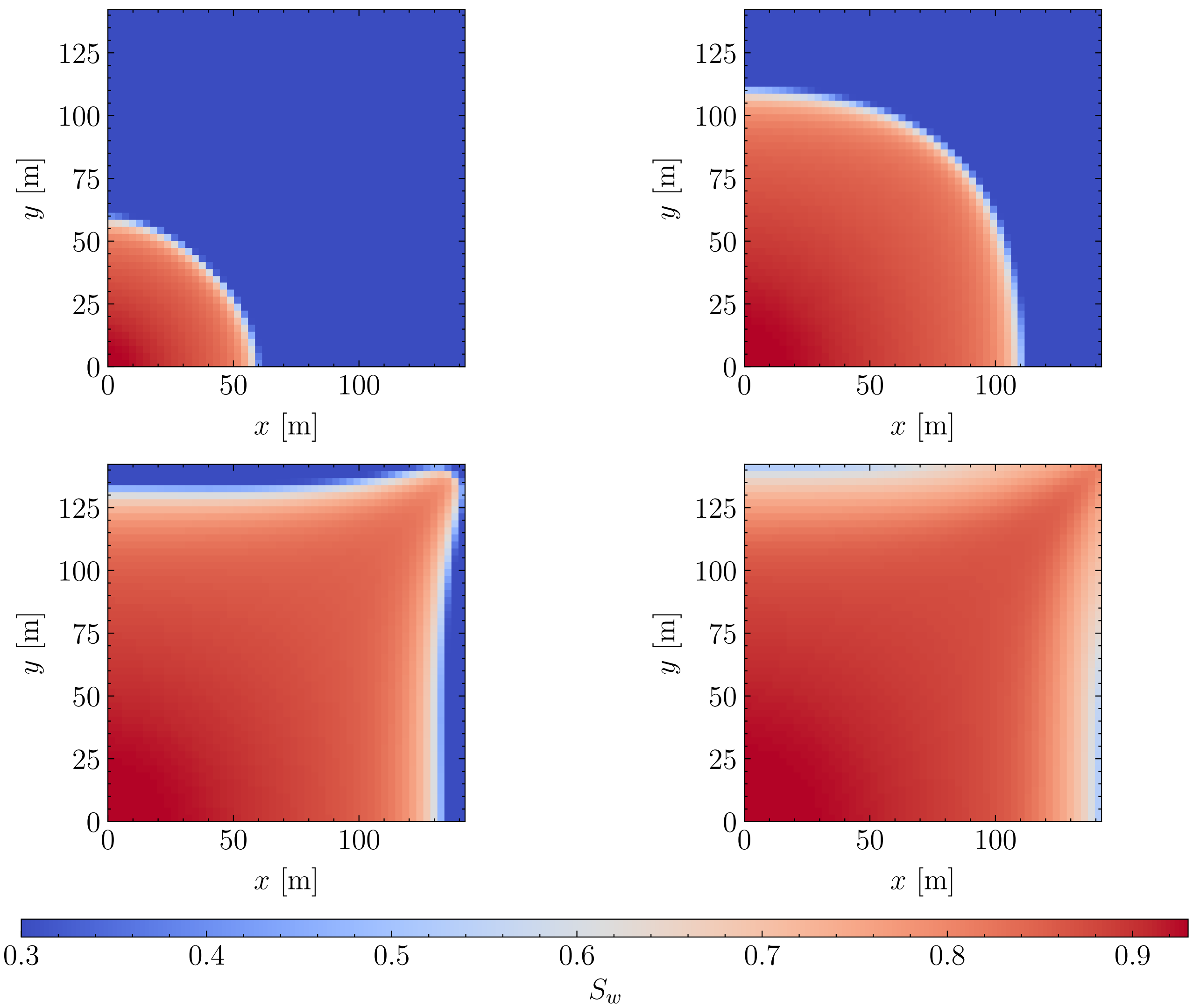


Figure 12: Spatial distribution of water saturation within the domain at four distinct time instants.

References

[1] Abreu, E. (2014). Numerical modelling of three-phase immiscible flow in heterogeneous porous media with gravitational effects. *Mathematics and Computers in Simulation*, 97:234–259.

[2] Abreu, E., Furtado, F., and Pereira, F. (2004). On the numerical simulation of three-phase reservoir transport problems. *Transport Theory and Statistical Physics*, 33(5-7):503–526.

[3] Blunt, M. J. (2017). *Multiphase Flow in Permeable Media: A Pore-Scale Perspective*. Cambridge University Press, Cambridge.

[4] Brooks, R. and Corey, T. (1964). Hydraulic properties of porous media. *Hydrology Papers, Colorado State University*, 24:37.

[5] Carrillo, F. J., Bourg, I. C., and Soulaine, C. (2020). Multiphase flow modeling in multiscale porous media: An open-source micro-continuum approach. *Journal of Computational Physics: X*, 8:100073.

[6] Chen, Z., Huan, G., and Ma, Y. (2006). *Computational methods for multiphase flows in porous media*. SIAM.

[7] Corey, A., Rathjens, C., Henderson, J., and Wyllie, M. (1956). Three-phase relative permeability. *Journal of Petroleum Technology*, 8(11):63–65.

[8] de Abreu, E. C. (2007). *Modelagem e simulação computacional de escoamentos trifásicos em reservatórios de petróleo heterogêneos*. Tese (doutorado), Universidade do Estado do Rio de Janeiro, Nova Friburgo, RJ, Brasil. Programa de Pós-Graduação em Modelagem Computacional.

[9] Durlofsky, L. J. (1993). A triangle based mixed finite element—finite volume technique for modeling two phase flow through porous media. *Journal of Computational Physics*, 105(2):252–266.

[10] Gaucher, D. and Lindley, D. (1960). Waterflood performance in a stratified, five-spot reservoir - a scaled-model study. *Transactions of the AIME*, 219(01):208–215.

[11] Horgue, P., Soulaine, C., Franc, J., Guibert, R., and Debenest, G. (2015). An open-source toolbox for multiphase flow in porous media. *Computer Physics Communications*, 187:217–226.

[12] Leverett, M. and Lewis, W. (1941). Steady flow of gas-oil-water mixtures through unconsolidated sands. *Transactions of the AIME*, 142(01):107–116.

[13] Panday, S., Wu, Y., Huyakorn, P., and Springer, E. (1994). A three-dimensional multiphase flow model for assessing NAPL contamination in porous and fractured media, 2. porous medium simulation examples. *Journal of Contaminant Hydrology*, 16(2):131–156.

[14] Peaceman, D. (1978). Interpretation of well-block pressures in numerical reservoir simulation(includes associated paper 6988). *Society of Petroleum Engineers Journal*, 18(03):183–194.

[15] Quinelato, T. O., de Paula, F. F., Igreja, I., et al. (2022). On the injectivity estimation in foam EOR. *Journal of Petroleum Exploration and Production Technology*, 12:2723–2734.

[16] van Leer, B. (1974). Towards the ultimate conservative difference scheme. ii. monotonicity and conservation combined in a second-order scheme. *Journal of Computational Physics*, 14(4):361–370.